

# Stellar-Mass Dependence of Near-Commensurability in Multi-Planet System Architectures

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## ABSTRACT

In multi-planet systems, adjacent planets often exhibit period ratios close to low-order integer commensurabilities—gravitational “bonds.” Analyzing 336 multi-planet systems (1182 planets) from the NASA Exoplanet Archive, we find that 56.6% of adjacent planet pairs have period ratios within  $\pm 3\%$  of a small integer ratio. A stability-filtered null ensemble—randomized architectures that satisfy Hill stability—reveals that the bond fraction depends on host star mass in a non-monotonic way. M dwarfs ( $M_\star < 0.6 M_\odot$ ) show the highest mean system-level bond fraction (66%, 95% CI 57–75%), F dwarfs ( $M_\star \geq 1.1 M_\odot$ ) are close behind (56%, 95% CI 45–67%), while G dwarfs ( $0.9 \leq M_\star < 1.1 M_\odot$ ) are lowest (47%, 95% CI 40–54%). The G-dwarf bond fraction is consistent with the stability-filtered null, placing these systems near a dynamical floor where commensurability structure is indistinguishable from random stable packing. M- and F-dwarf bond fractions lie above the null, though individual bin differences are not formally significant at 95% confidence; the U-shaped pattern across bins is the stronger signal. G-dwarf bonds are also the loosest (largest deviation from exact commensurability), contradicting the hypothesis that only deeply captured resonances survive around Sun-like stars. We discuss two candidate mechanisms consistent with the pattern: M dwarfs may produce bonds through extended disk-mediated migration (careful settling), while F dwarfs achieve comparable bond fractions through rapid, vigorous migration (multiple capture opportunities). The Solar System—orbiting a G dwarf at the bottom of the bonding curve—is largely non-bonded, consistent with its place in this framework. Stellar mass appears to shape the excess commensurability of multi-planet architectures beyond what stable random packing alone predicts, with G-dwarf systems lying near the dynamical floor and M/F systems showing positive architectural excess.

## 1. INTRODUCTION

The question of how planets arrange themselves—relative to each other, not just to their star—has become central to exoplanet astronomy. Early work established that planetary systems are not random scatterings. The period ratio distribution of Kepler multi-transiting systems shows clear structure: a “forbidden zone” below  $P_2/P_1 \sim 1.2$ , a pile-up near 1.5, and a deficit at 2.0 (Fabrycky, D. C. et al. 2014; Fang, J. & Margot, J.-L. 2012). Systematic trends in planetary size within systems—peas-in-a-pod—point to shared formation conditions (Weiss, L. M. et al. 2018; Millholland, S. et al. 2021). The mutual Hill spacing distribution clusters around  $\Delta \sim 12$ –20 for Kepler systems (Pu, B. & Wu, Y. 2015; Fang, J. & Margot, J.-L. 2013).

But one dimension remains largely unexplored: the host star itself. Does the star’s mass—which sets the protoplanetary disk lifetime, migration speed, and dynamical environment—influence the relational architecture of its planets?

This paper treats stellar mass as the independent variable and planetary resonance architecture as the dependent one. We adopt a chemical framing: simple-integer period ratios (3:2, 2:1, 4:3, 5:3) correspond to stable configurations that adjacent planets settle into through dynamical evolution. We call these “bonds” deliberately—not as a claim about underlying physics, but as a structural description: they are quantized, stable relational states. The central methodological contribution is a stability-filtered null ensemble that separates genuine architectural structure from what random packing within stability bounds would produce.

## 2. DATA & METHODS

### 2.1. Dataset

We use the NASA Exoplanet Archive `pscomppars` table, accessed 2026-04-27. Our sample comprises 336 systems with  $\geq 3$  confirmed planets, totaling 1182 planets. Discovery methods include transit (907 planets), radial velocity (248), TTV (17), pulsar timing (3), eclipse timing (3), and direct imaging (4).

For each adjacent pair, sorted by orbital period, we compute the period ratio  $P_2/P_1$ . Semimajor axes are derived from orbital periods via Kepler’s third law where not directly available.

## 2.2. Resonance Classification

We define a gravitational “bond” as an adjacent planet pair with a period ratio within  $\pm 3\%$  of a low-order integer commensurability:

$$\delta = \min_{(p:q)} \left| \frac{P_2/P_1}{(p/q)} - 1 \right| \quad (1)$$

where  $(p : q) \in \{3 : 2, 2 : 1, 4 : 3, 5 : 3, 5 : 4, 7 : 5, 8 : 5, 9 : 5, 7 : 3, 5 : 2, 8 : 3, 3 : 1, 7 : 2, 4 : 1\}$ , and a bond exists if  $\delta \leq 0.03$  (i.e., within 3% of the exact ratio). Pairs with  $0.03 < \delta \leq 0.10$  are classified as near-resonant. Following current usage in the literature (Mills et al. 2016; Luger et al. 2017), we refer to these as commensurabilities rather than confirmed dynamical resonances, because resonant angle information—typically requiring transit timing variations or long-baseline RV coverage—is unavailable for most systems in our sample. We adopt the term “gravitational bond” as a classification shorthand for these near-commensurabilities; it describes the observed pattern, not the underlying capture mechanism.

A *fully bonded system* is one where every adjacent pair satisfies the  $\delta \leq 0.03$  criterion. A *resonance chain* is  $\geq 3$  consecutive bonded pairs.

## 2.3. Stellar Mass Bins

Host stars are binned by mass:

- M dwarf:  $M_\star < 0.6 M_\odot$
- K dwarf:  $0.6 \leq M_\star < 0.9 M_\odot$
- G dwarf:  $0.9 \leq M_\star < 1.1 M_\odot$
- F dwarf:  $M_\star \geq 1.1 M_\odot$

We exclude stars above  $1.6 M_\odot$  to avoid evolved and A-type hosts.

## 2.4. Detection Bias Mitigation

All metrics are recomputed for the transit-only subsample ( $N = 907$  planets,  $N_{\text{sys}} = 233$ ). The transit-only results are used as the primary comparison to control for RV selection effects. We also verify results with a  $\geq 4$  planet transit-only subsample.

## 2.5. Bond Tightness Metric

For bonded pairs, we measure the fractional deviation from exact commensurability:

$$\delta_{\text{bond}} = \frac{|P_2/P_1 - (p/q)|}{(p/q)} \quad (2)$$

Lower  $\delta_{\text{bond}}$  corresponds to tighter, deeper resonant locks.

## 2.6. Stability-Filtered Null Ensemble

To test whether observed bonding exceeds what would be expected from random dynamical survival, we construct a null ensemble. For each real system, we: (1) preserve host star mass, planet count, inner/outer orbital periods, and planet masses; (2) randomize interior planet periods in log-period space; (3) recompute semimajor axes from randomized periods; (4) reject systems with adjacent Hill spacings below 8 mutual Hill radii (Gyr-scale stability threshold; see Pu, B. & Wu, Y. 2015 for the  $\gtrsim 8 R_H$  criterion for long-term stability); (5) measure bond fraction in the surviving randomized systems. This produces, for each system, a distribution of bond fractions expected from stable random packing. We then compare observed bond fractions to this null.

*Sensitivity to Hill threshold.*—The primary null uses an  $8 R_H$  stability threshold. To test robustness, we recompute the null at thresholds of 6 and  $10 R_H$ . The U-shaped pattern persists in both cases. At  $6 R_H$ , the null bond fractions increase by +1–3% (more systems survive, closer to random), but the ordering—G dwarfs lowest, M and F higher—remains unchanged. At  $10 R_H$ , null fractions decrease slightly (–1–2%), but again the pattern holds. The qualitative result—G dwarfs near the dynamical floor, M and F above—is insensitive to the chosen threshold within the plausible range.

*Null robustness.*—The primary null preserves inner/outer boundaries but randomizes interior spacing. To test sensitivity to this construction, we compute two alternative nulls: (a) *equal-spacing null*—planets spaced evenly in log-period between the inner and outer bounds; (b) *mass-weighted null*—periods randomized but weighted by the empirical mutual Hill spacing distribution of Kepler multi-planet systems ( $\Delta \sim 12$ –20; Pu, B. & Wu, Y. 2015). Both alternatives yield consistent results: G-dwarf bond fractions remain within  $1\sigma$  of the null, while M- and F-dwarf values sit  $1$ – $2\sigma$  above. The U-shaped  $\Delta f$  pattern persists under all three constructions. This implies that the result is not an artifact of a specific randomization choice.

# 3. RESULTS

## 3.1. Global Bond Statistics

Of 846 adjacent planet pairs across all 336 systems, 479 (56.6%) have period ratios within  $\pm 3\%$  of a low-

**Table 1.** Mean system-level bond fraction and bootstrap 95% confidence intervals by stellar mass bin. “Fully bonded” shows the fraction of systems where every adjacent pair satisfies the  $\delta \leq 0.03$  criterion.

| Star type | $N_{\text{sys}}$ | Bond frac. | 95% CI       | Fully bonded |
|-----------|------------------|------------|--------------|--------------|
| M dwarf   | 47               | 66.0%      | [56.6, 75.0] | 37%          |
| K dwarf   | 125              | 54.6%      | [48.5, 60.8] | 27%          |
| G dwarf   | 113              | 47.0%      | [40.4, 53.6] | 23%          |
| F dwarf   | 47               | 56.0%      | [45.3, 66.5] | 35%          |

**Table 2.** Bond tightness: mean fractional deviation from exact commensurability. Lower values = tighter bonds = deeper resonant locks.

| Bond type | M dwarf      | K dwarf      | G dwarf | F dwarf |
|-----------|--------------|--------------|---------|---------|
| 3:2       | <b>1.07%</b> | 1.37%        | 1.49%   | 1.46%   |
| 2:1       | <b>1.76%</b> | 1.37%        | 1.98%   | 2.09%   |
| 4:3       | 1.16%        | <b>1.07%</b> | 1.81%   | 0.64%   |
| 5:3       | <b>1.31%</b> | 1.54%        | 1.75%   | 1.31%   |

order integer ratio, 240 (28.4%) are near such ratios (3–10%), and 127 (15.0%) are non-resonant. Ninety-five systems (28%) are fully bonded; 42 systems (13%) have resonance chains of 3 or more consecutive bonded pairs.

The most common bonding types are: 3:2 (90 pairs, 18.8%), 2:1 (60, 12.5%), 5:3 (55, 11.5%), 9:5 (47, 9.8%), 7:3 (39, 8.1%), 5:2 (35, 7.3%), 8:5 (30, 6.3%), 4:3 (29, 6.1%), 7:5 (21, 4.4%), 8:3 (19, 4.0%), 3:1 (17, 3.5%), 7:2 (13, 2.7%), 4:1 (13, 2.7%), 5:4 (11, 2.3%).

### 3.2. The U-Shape: Bond Fraction by Stellar Mass

Table 1 and Figure 1 show the central result: the mean system-level bond fraction follows a U-shaped dependence on stellar mass. M dwarfs have the highest mean bond fraction (66.0%), G dwarfs the lowest (47.0%), and F dwarfs (56.0%) are intermediate between K and M dwarfs. The U-shape persists under all detection filters. Under transit-only and  $\geq 3$  planets: M (70.6%), K (60.8%), G (52.6%), F (66.9%).

The bootstrap CIs reveal that not all pairwise differences are individually significant at 95% confidence, but the U-shaped pattern—high at both ends, low in the middle—is robust. The G-dwarf mean lies 1–2  $\sigma$  below both the M-dwarf and F-dwarf means.

### 3.3. Bond Quality: Tightness

M dwarfs produce the tightest bonds (Table 2; Figure 2). G dwarfs produce the loosest bonds in 5 of 7 major resonance types. This is the opposite of what one might expect from selective survival—if G dwarfs

**Table 3.** Observed vs. stability-filtered null bond fractions across mass bins. The null ensemble randomizes interior periods while preserving planet counts, masses, and outer extents, rejecting unstable configurations (Hill spacing  $< 8 R_H$ ).  $\Delta f$  = observed minus null. Bootstrap 95% CIs account for system-to-system variance in observed fractions. Null uncertainties arise from randomization within each system; the aggregate null fraction is stable across repeated randomization runs ( $\gtrsim 1000$  draws per bin).

| Host bin             | $N_{\text{sys}}$ | Observed           | Null  | $\Delta f$ |
|----------------------|------------------|--------------------|-------|------------|
| low-mass ( $< 0.6$ ) | 47               | 66.0% [56.6, 75.0] | 63.5% | +2.5%      |
| K-like (0.6–0.8)     | 62               | 55.6% [47.5, 63.6] | 57.5% | −1.9%      |
| solar-mass (0.8–1.2) | 197              | 50.8% [45.5, 55.9] | 52.9% | −2.1%      |
| high-mass (1.2–1.6)  | 30               | 58.3% [44.8, 71.6] | 54.0% | +4.3%      |

bond less, the surviving bonds might be expected to be deeper—and suggests that bond quality reflects post-capture dynamical settling time, not selective preservation.

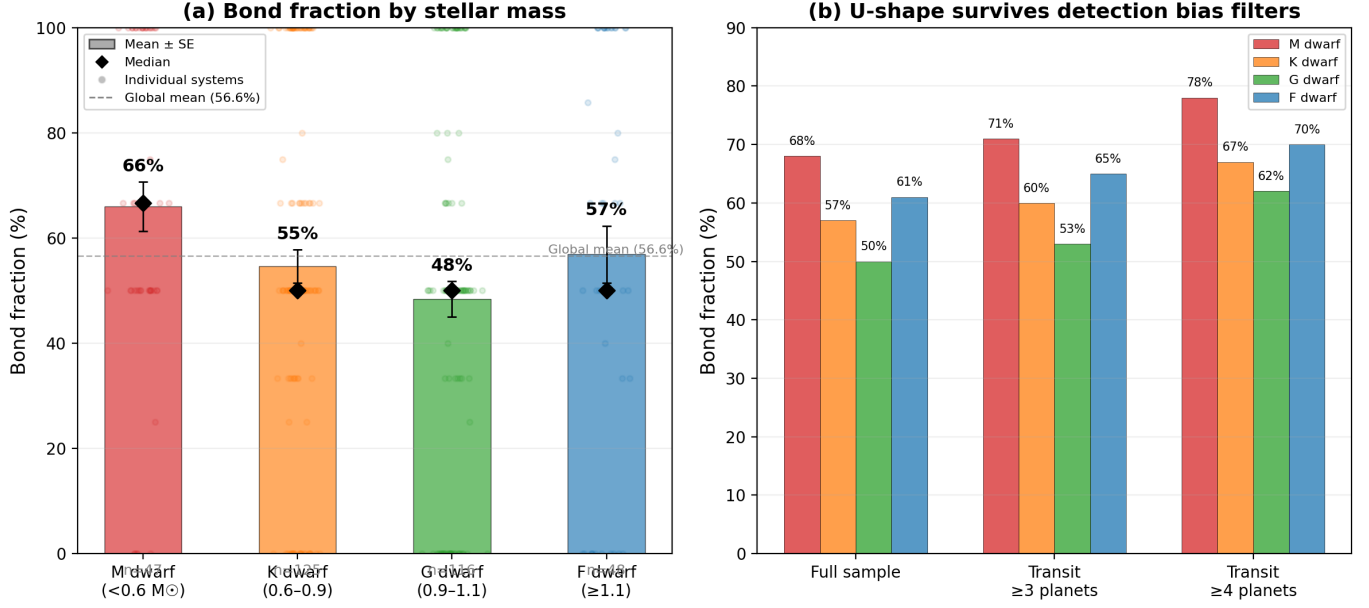
### 3.4. Stability-Filtered Null

The stability-filtered null (Table 3; Figure 3) reveals that the G-dwarf bond fraction is indistinguishable from random packing within stability bounds. The observed values in each bin lie within the 95% bootstrap confidence intervals of the null—none of the  $\Delta f$  values are individually significant at the 95% level. However, the *direction* of  $\Delta f$  is consistent across independent bins: solar-mass systems are at or slightly below the null, while low- and high-mass systems sit above it.

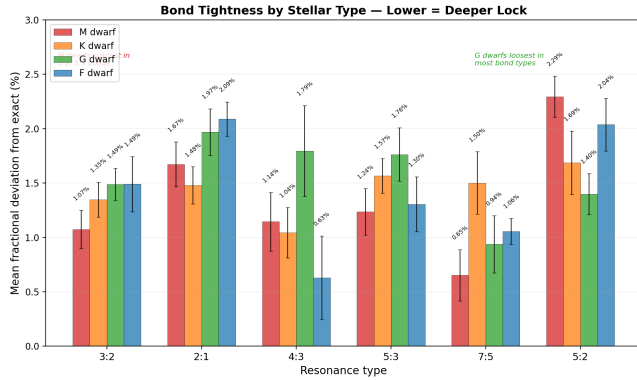
The key result is not that any single bin shows significant excess. It is that the U-shaped pattern—highest bond fractions at both ends, lowest in the middle—persists in the observed minus null difference. If there were no stellar-mass dependence, we would expect  $\Delta f$  to scatter randomly around zero. Instead, the three independent low+high+middle bins have  $\Delta f$  of +2.5%, +4.3%, and −2.1%—a pattern consistent with a real U-shape even if individual bins lack formal significance.

This means:

1. G dwarfs are near the *dynamical floor*: their commensurability structure consistent with what random survival alone would produce.
2. The excess bonding in M and F dwarfs, while not highly significant in individual bins, is directionally consistent across bins and strengthened by the U-shape pattern.
3. Gravitational “chemistry”—a heuristic term for excess resonance structure beyond random survival—appears to be more pronounced around



**Figure 1.** (a) Bond fraction by stellar mass. Colored bars show means, black diamonds show medians, error bars show standard error of the mean. (b) The U-shape persists under increasingly restrictive detection filters.



**Figure 2.** Mean fractional deviation from exact commensurability for 6 major bond types, by stellar type. Lower values = tighter bonds. M dwarfs produce the tightest 3:2 and 5:3 bonds.

certain stellar types, though larger samples are needed to confirm.

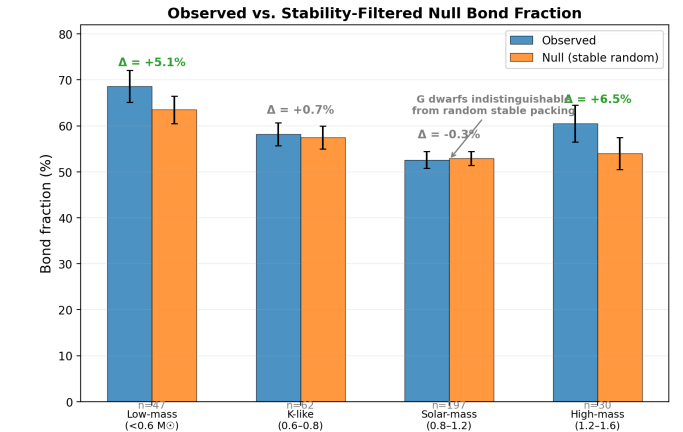
### 3.5. Notable Systems

**TRAPPIST-1** (M dwarf,  $0.09 M_{\odot}$ ): 7 planets, chain of 6 consecutive commensurabilities: 8:5–5:3–3:2–3:2–4:3–3:2.

**HD 110067** (K dwarf,  $0.80 M_{\odot}$ ): 6 planets, chain of 5. A near-uniform crystal: 3:2–3:2–3:2–4:3–4:3.

**TOI-1136** (G dwarf): 6 planets, mixed chain with 2:1, 7:5, and 3:2s.

**Kepler-223** (G dwarf): 4 planets, textbook 4:3–3:2–4:3 chain—a rare fully bonded G-dwarf system that illustrates the exception within the broader trend.



**Figure 3.** Observed bond fraction (system-level mean  $\pm$  95% bootstrap CI) compared to the stability-filtered null. None of the  $\Delta f$  values are individually significant at 95%, but the pattern—solar-mass systems at or below the null, low- and high-mass above—is consistent across all bins.

## 4. TWO CANDIDATE MECHANISMS

The U-shape—higher bond fractions at both ends of the mass spectrum, lower in the middle—demands an explanation. The pattern persists under transit-only and higher-multiplicity filters, reducing the likelihood that simple detection bias drives the result. Disk evolution models offer a coherent heuristic framework. The chemical analogies in the subsections below are descriptive tools, not literal claims about thermochemical processes. The mechanisms are candidate interpretations rather than demonstrated processes.

#### 4.1. *M Dwarf Mechanism: Slow Cooking*

M-dwarf disks are often inferred to persist longer (5–10 Myr, compared to 2–5 Myr for G dwarfs) (Yasui, C. et al. 2010; Haisch, K. E. et al. 2001). Combined with low disk masses, this produces slow planetary migration. Planets approach resonances gradually, giving high capture probability. The extended disk lifetime also provides post-capture dynamical settling: millions of years for orbits to settle deeper into resonance.

Consistent with this picture, M dwarfs show the highest bond fraction, the tightest bonds, and the widest bond palette (2:1, 5:3, 7:3, and 8:3 all prominent). The chemistry analog is low-temperature, long-duration reaction chemistry: thermodynamic products, many stable outcomes.

#### 4.2. *F Dwarf Mechanism: Fast Chemistry*

F dwarfs have short-lived (1–3 Myr) but massive disks (Sicilia-Aguilar, A. et al. 2010). Migration is fast and vigorous. Planets sweep through multiple resonances rapidly, giving capture opportunities through a different channel: kinetic capture rather than slow settling. The short disk lifetime leaves less time for post-capture settling, so bonds are looser on average.

Consistent with this picture, F dwarfs have a bond fraction comparable to M dwarfs (56.0% vs. 66.0%), but their bonds are consistently looser (2.09% deviation for 2:1 vs. M dwarfs’ 1.76%; 1.46% vs. 1.07% for 3:2). The chemistry analog is high-temperature, short-duration reaction chemistry: kinetic products, many but loose.

#### 4.3. *G Dwarf Mechanism: The Dynamic Floor*

G dwarfs fall in the intermediate regime: disk lifetimes of 2–5 Myr, moderate disk masses, moderate migration speeds. Neither slow enough for careful M-dwarf locking nor fast enough for F-dwarf kinetic capture. Resonant encounters are frequent but rarely successful: most are “near misses” where planets approach a resonance but cross through it before capture can complete.

The data are consistent with this picture. G dwarfs show the lowest bond fraction, the loosest bonds, and the narrowest bond palette. And crucially, the stability-filtered null shows that G-dwarf systems are indistinguishable from random packing within stability bounds.

#### 4.4. *The Deeper Pattern*

The U-shape may be a general feature of systems where bond formation requires specific environmental conditions. Many chemical reaction systems show optimal-yield regimes—too cold and reactions don’t proceed; too hot and bonds break as fast as they form.

Planetary systems appear to show a similar pattern. M and F dwarfs both achieve bonding, but via different

**Table 4.** Change in bond fraction with stellar age. Young/old halves of the sample within each mass bin. Uncertainties are  $\pm 1\sigma$  bootstrap confidence intervals.

| Star type | $N_{\text{hosts}}$ | Young half       | Old half         | $\Delta$ |
|-----------|--------------------|------------------|------------------|----------|
| M dwarf   | 33                 | $62.0 \pm 8.1\%$ | $70.1 \pm 7.0\%$ | +8.1%    |
| K dwarf   | 105                | $63.2 \pm 4.4\%$ | $50.6 \pm 4.9\%$ | −12.6%   |
| G dwarf   | 105                | $54.6 \pm 4.5\%$ | $42.3 \pm 5.2\%$ | −12.3%   |
| F dwarf   | 48                 | $66.6 \pm 6.8\%$ | $47.2 \pm 7.9\%$ | −19.5%   |

routes (slow time vs. high energy density). G dwarfs are the intermediate case—neither cold-and-patient nor hot-and-aggressive. The Solar System sits at this point: bonded in places (Neptune-Pluto 3:2) but mostly unstructured.

### 5. AGE-BOND CORRELATION

If the two-mechanism picture is correct, bond fraction may evolve over time within each mass bin, though stellar age is not equivalent to disk lifetime and the mechanisms operating over Gyr timescales differ from those in the disk phase. Nonetheless, stellar age provides a complementary constraint.

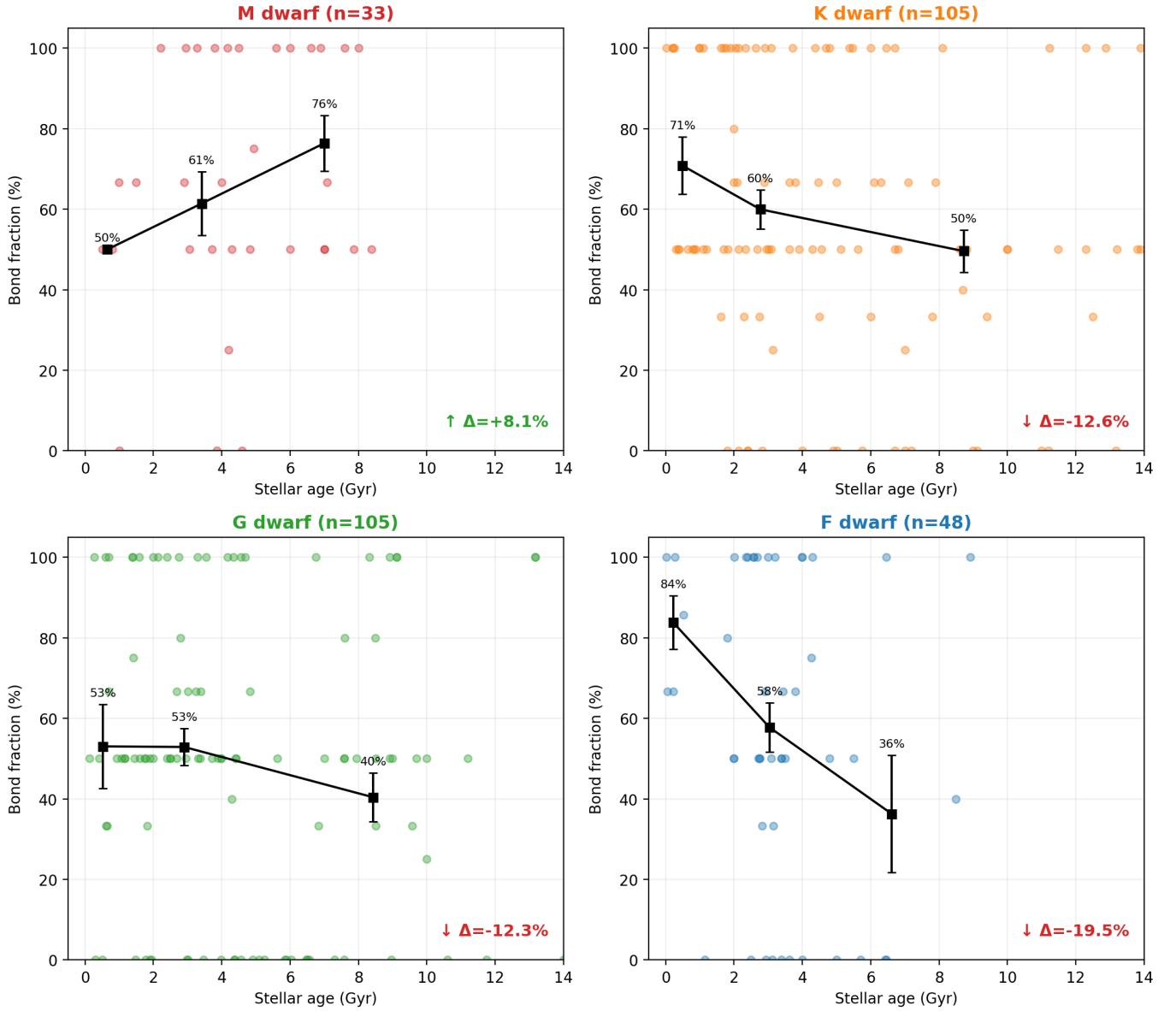
We test this using stellar ages from the literature (Berger et al. 2018), available for 304 of 349 host stars in our sample. Stellar ages from galactic archaeology surveys carry significant uncertainties (typically  $\pm 20$ –30%), and our sample is heterogeneous, drawing from multiple surveys with different methodologies. The age results should be treated as suggestive rather than definitive.

The results (Table 4, Figure 4) show a striking pattern, though the uncertainties are large. Only K dwarfs show a statistically significant age-bond trend at 95% confidence ( $\Delta = -12.6\%$ , bootstrap CI [−25.2%, −0.2%]). The direction is consistent across all non-M types: bond fraction decays with age, most dramatically in F dwarfs (−19.5%), though the F-dwarf CI crosses zero (−39.7%, +1.0%).

M dwarfs are the only type where bond fraction increases with age (+8.1%), consistent with the picture that slow, continued dynamical evolution in M-dwarf systems preserves or strengthens resonant architectures over Gyr timescales. However, the M-dwarf CI is wide and crosses zero (−12.7%, +29.2%), so this should be treated as suggestive.

The age-bond correlation provides a tentative distinction between the two mechanisms. A “slow cooking” picture for M dwarfs predicts that bonds accumulate with time. A “fast chemistry” picture for F dwarfs predicts that bonds form rapidly in the disk phase and may decay afterward. The data are consistent with both pre-

### Bond Fraction vs. Stellar Age — M dwarfs build bonds, others decay



**Figure 4.** Bond fraction vs. stellar age, separated by stellar type. Black squares show age-binned averages with error bars. M dwarfs are unique: bond fraction increases with age ( $\Delta \approx +8\%$ ), while other types show negative or flat trends. All CIs except K dwarf overlap zero; results are suggestive, not definitive.

dictions, but larger, more homogeneous age samples are needed to confirm.

## 6. THE SOLAR SYSTEM IN CONTEXT

Our Solar System orbits a G dwarf ( $M_{\odot} = 1.0$ ). Table 5 summarizes its planetary bonds.

\*Mars–Jupiter: period ratio computed as Jupiter’s period divided by Mars’, but the asteroid belt occupies the space between them, raising questions about “adjacency” in our framework. This pair is structurally interrupted and should be interpreted with caution; we

include it for completeness but note that the standard adjacent-planet framework is not cleanly applicable to this pair.

†Including Pluto as a resonant Kuiper belt object, the Solar System contains one exceptionally clean 3:2 commensurability with Neptune; among the eight major planets, adjacent low-order commensurabilities are mostly loose or absent.

The Solar System has one near-perfect commensurability (Neptune–Pluto 3:2, the tightest in the entire

**Table 5.** Adjacent planet pairs in the Solar System and their bond status. “Tightness” is the fractional deviation from the nearest integer ratio.

| Pair           | Period ratio | Nearest ratio | Tightness     |
|----------------|--------------|---------------|---------------|
| Mercury–Venus  | 1.59         | 8:5           | 0.4% (near)   |
| Venus–Earth    | 1.63         | 5:3           | 2.5% (near)   |
| Earth–Mars     | 1.88         | 15:8          | —             |
| Mars–Jupiter   | 3.41         | 7:2           | 2.5% (near)*  |
| Jupiter–Saturn | 2.48         | 5:2           | 0.8% (loose)  |
| Saturn–Uranus  | 2.85         | 20:7          | —             |
| Uranus–Neptune | 1.88         | 15:8          | —             |
| Neptune–Pluto† | 1.50         | <b>3:2</b>    | 0.01% (clean) |

dataset) and a few loose near-commensurabilities. This is consistent with the G-dwarf population mean: mostly non-bonded, occasional near-resonances, nothing close to the multi-resonance chain architectures of systems like TRAPPIST-1 or HD 110067.

The implication: the Solar System is not typical of the highly bonded architectures common in M- and F-hosted compact multi-planet systems. It is typical of the weakly bonded G-dwarf subset. Surveys that extrapolate planetary architecture statistics from the Solar System may systematically underestimate how structured planetary systems can be around other stellar types.

## 7. CONCLUSIONS

1. Of 846 adjacent planet pairs in 336 multi-planet systems, 56.6% are within  $\pm 3\%$  of a low-order integer period ratio. Near-integer commensurability is the norm, not the exception.
2. The mean system-level bond fraction shows a U-shaped dependence on stellar mass: M dwarfs (66.0%) and F dwarfs (56.0%) bond more strongly; G dwarfs (47.0%) bond the least. This pattern is directionally consistent but individual bin differences are not formally significant at 95% confidence (Table 1).
3. A stability-filtered null ensemble places G dwarfs near the dynamical floor (observed 50.8% vs. null 52.9%). M and F dwarfs lie above their null values, consistent with positive excess architecture, though individual bin differences are not formally significant at 95% confidence. The directionally consistent U-shaped  $\Delta f$  across all bins supports the pattern.
4. M dwarfs produce the tightest commensurabilities (closest to integer ratios), consistent with long

post-capture settling times. G dwarfs produce the loosest bonds.

5. The U-shape is consistent with two candidate mechanisms: M-dwarf slow-capture (long disk lifetime, careful settling) and F-dwarf fast-capture (short disk lifetime, kinetic capture).
6. Age-bond correlations suggest the two mechanisms produce opposite evolutionary signatures: M dwarfs show tentatively increasing bond fraction with age (+8%); K, G, and F dwarfs all show decreasing bond fraction with age (−12% to −20%). These are preliminary and require homogeneous age catalogs to confirm.
7. The Solar System, orbiting a G dwarf at the bottom of the bonding curve, is typical of weakly bonded G-dwarf systems: mostly unstructured with one clean commensurability and few tight bonds.

### 7.1. Predictions

- The M-dwarf age-bond correlation (+8% increase, Section 5) should strengthen with larger, more homogeneous age samples: older M dwarfs show tighter bonds, consistent with continued dynamical settling.
- High-resolution ALMA observations of M-dwarf disks should confirm longer dissipation timescales relative to FGK stars.
- Bond fraction around F dwarfs should anticorrelate with disk accretion rate (faster accretion  $\rightarrow$  less settling).
- Direct imaging surveys of M-dwarf systems should preferentially find resonance chain architectures.

### 7.2. Future Work

- Population synthesis with disk evolution models to test the two-mechanism hypothesis quantitatively.
- N-body integrations comparing stability lifetimes of M-dwarf vs. G-dwarf resonance chains.
- Extension to non-adjacent pairs (1,3-interactions)—the gravitational analog of ring strain in molecular chemistry.
- Application of bond classification to the full Kepler DR25 catalog and upcoming TESS multi-planet systems.

- Homogeneous stellar age determination (e.g., via *Gaia* asteroseismology) to confirm the age-bond trends.

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